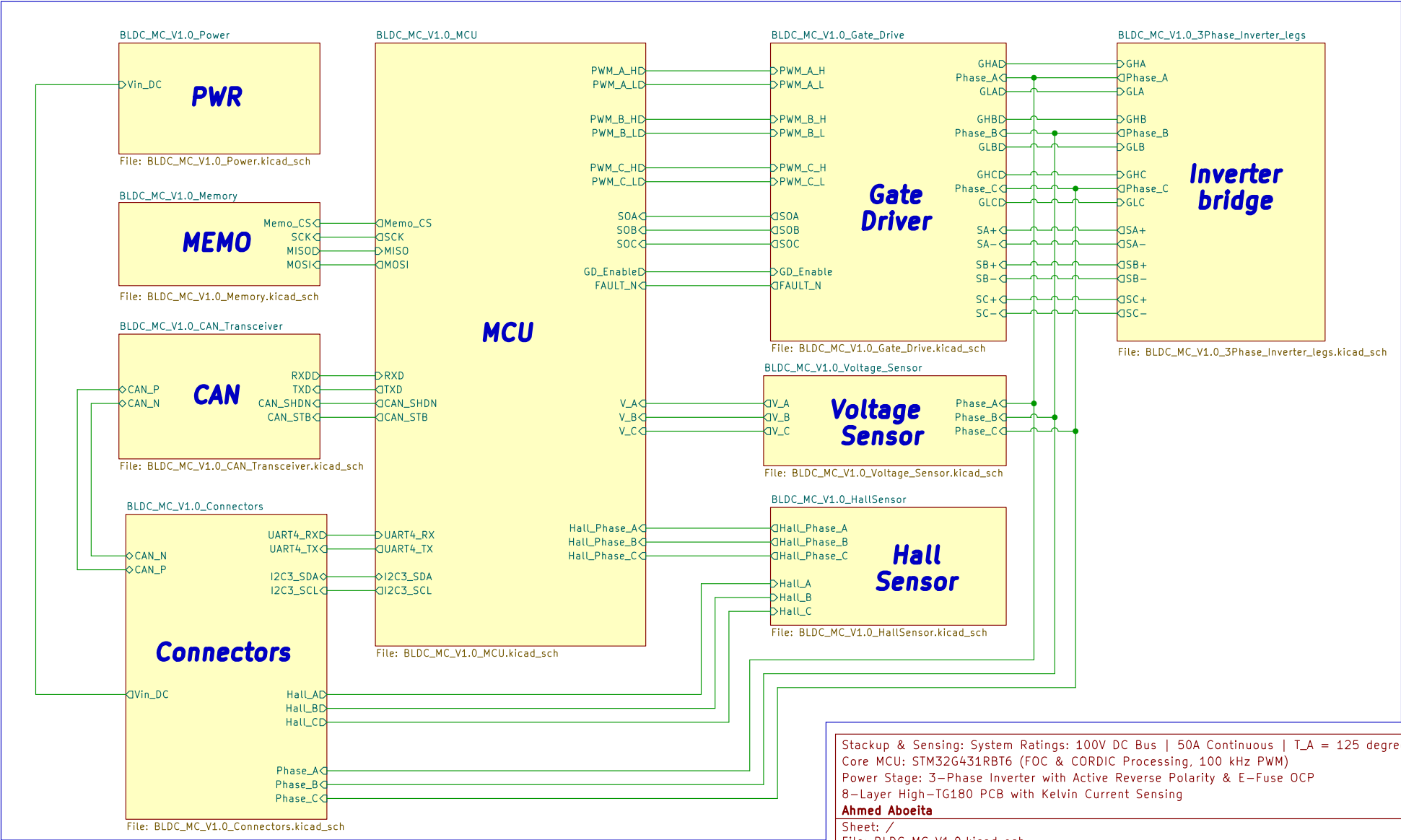


[1] BLDC_MC V1 60V 30A



Stackup & Sensing: System Ratings: 100V DC Bus | 50A Continuous | T_A = 125 degrees
Core MCU: STM32G431RBT6 (FOC & CORDIC Processing, 100 kHz PWM)
Power Stage: 3-Phase Inverter with Active Reverse Polarity & E-Fuse OCP
8-Layer High-TG180 PCB with Kelvin Current Sensing

Ahmed Aboeita

Sheet: /
File: BLDC_MC_V1.0.kicad_sch

Title: BLDC_MC

Size: A4 Date: 2026-08-17

KiCad E.D.A. 10.0.5

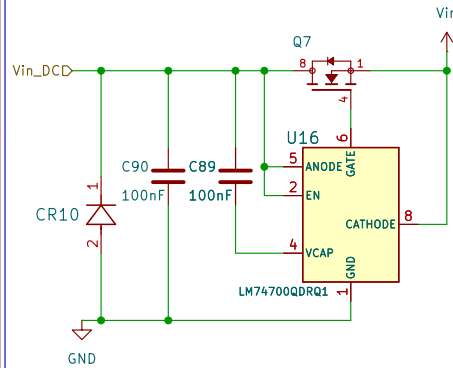
Rev: V1.0

Id: 1/10

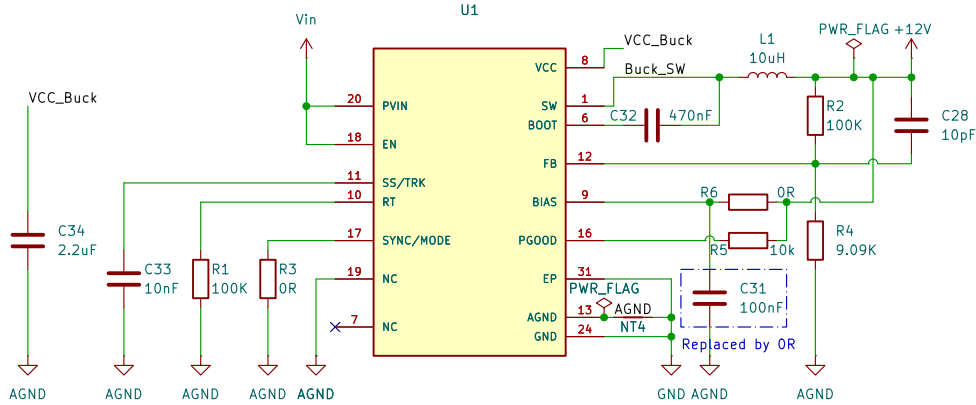
[1] Power Conversion Vin to 12V and 5V to 3.3V

The maximum input voltage of this 60V so we can use for BLDC motors up to 60V

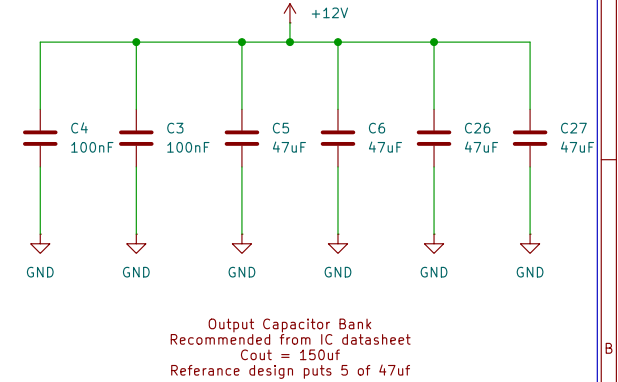
Input PWR Protection



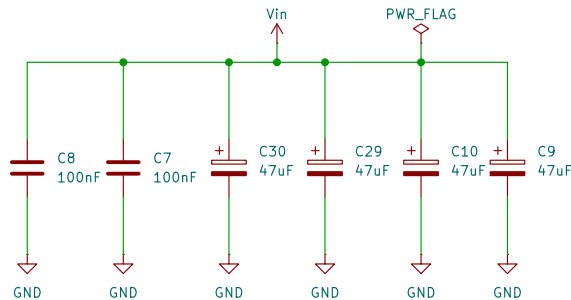
Buck IC Circuit



Output Cap Bank

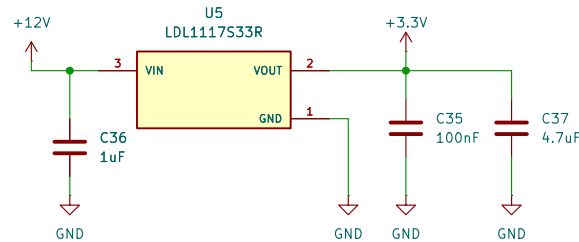


Input Cap Bank

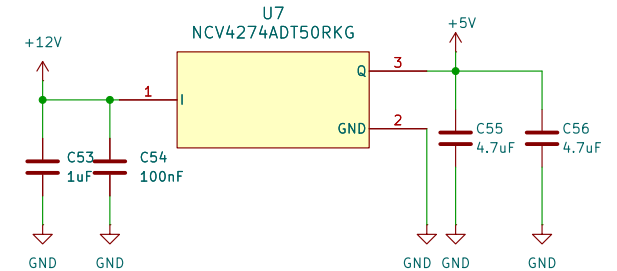


input Capacitor Bank
To make the input power pure DC

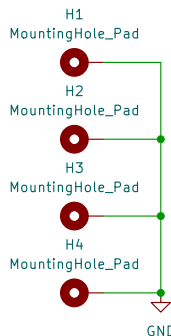
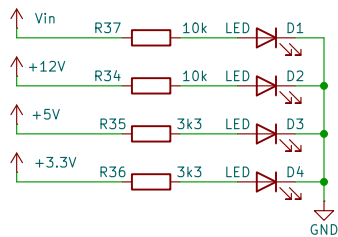
LDO Fixed out 3.3V



LDO Fixed out 5V



Indication Leds



Voltage Level	Usage
12V	Gate drive and MOSFET firing Voltage
5V	for Hall Sensor
3.3V	Controller and aux ICs

Stackup & Sensing: System Ratings: 100V DC Bus | 50A Continuous | T_A = 125 degrees
Core MCU: STM32G431RBT6 (FOC & CORDIC Processing, 100 kHz PWM)
Power Stage: 3-Phase Inverter with Active Reverse Polarity & E-Fuse OCP
8-Layer High-TG180 PCB with Kelvin Current Sensing

Ahmed Aboeita

Sheet: /BLDC_MC_V1.0_Power/
File: BLDC_MC_V1.0_Power.kicad_sch

Title: BLDC_MC

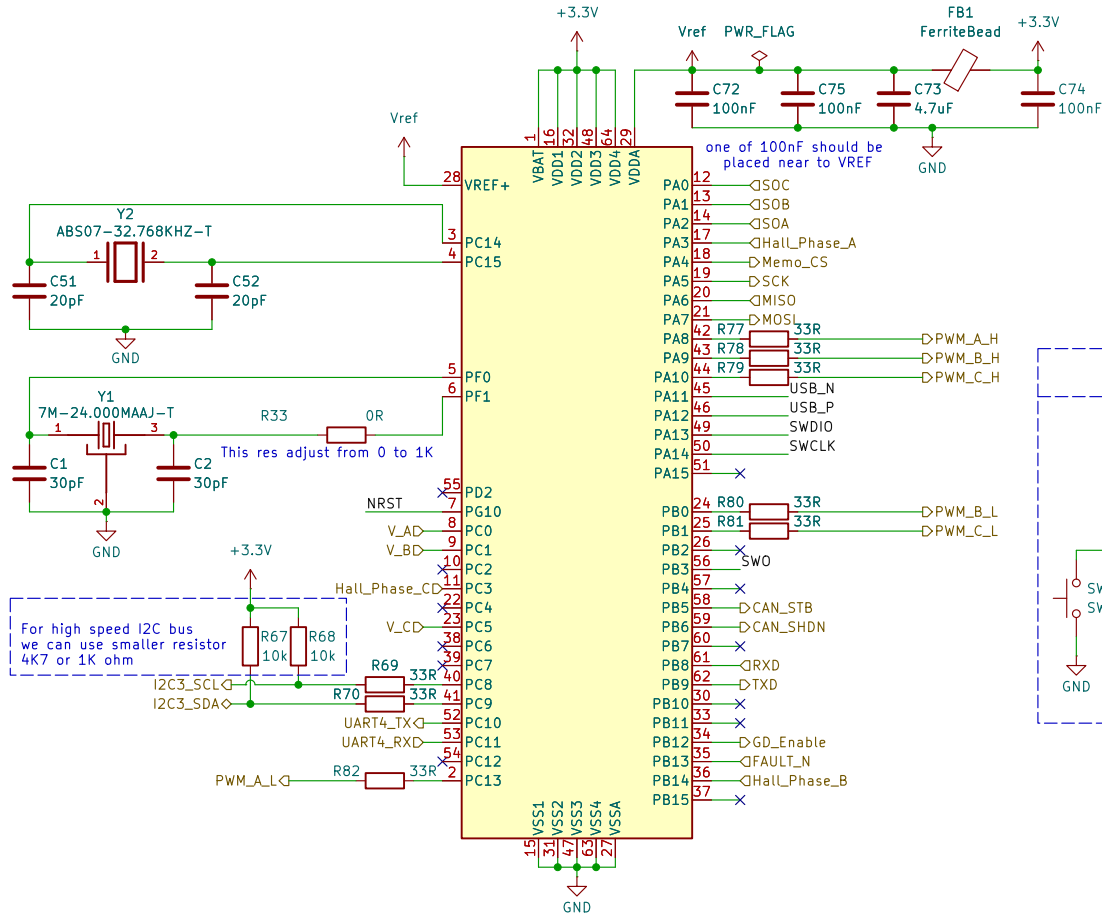
Size: A4
KiCad E.D.A. 10.0.5

Date: 2026-08-17

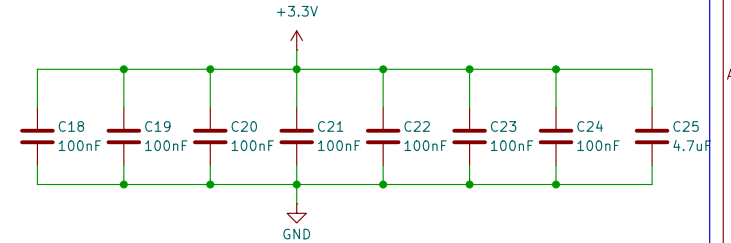
Rev: V1.0
Id: 2/10

[3] STM32G431RBT6 Controller Interface

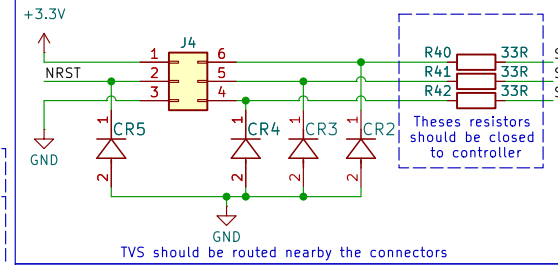
This MCU can handle the FOC control Algorithm



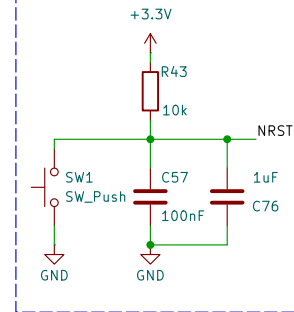
Power bypass Capacitors



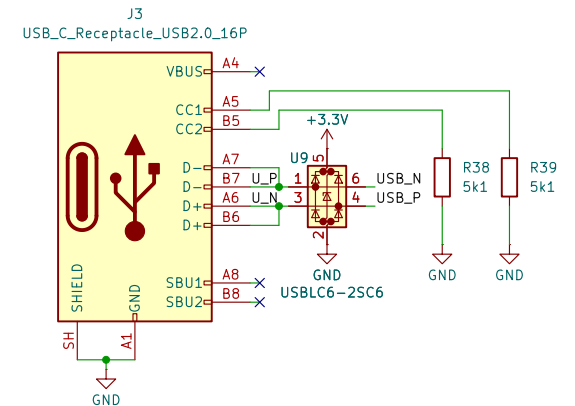
Debugger Interface (Serial Wire Debugger)



NRST circuit



USB-C



Stackup & Sensing: System Ratings: 100V DC Bus | 50A Continuous | T_A = 125 degrees
Core MCU: STM32G431RBT6 (FOC & CORDIC Processing, 100 kHz PWM)
Power Stage: 3-Phase Inverter with Active Reverse Polarity & E-Fuse OCP
8-Layer High-TG180 PCB with Kelvin Current Sensing

Ahmed Aboeita

Sheet: /BLDC_MC_V1.0_MCU/
File: BLDC_MC_V1.0_MCU.kicad_sch

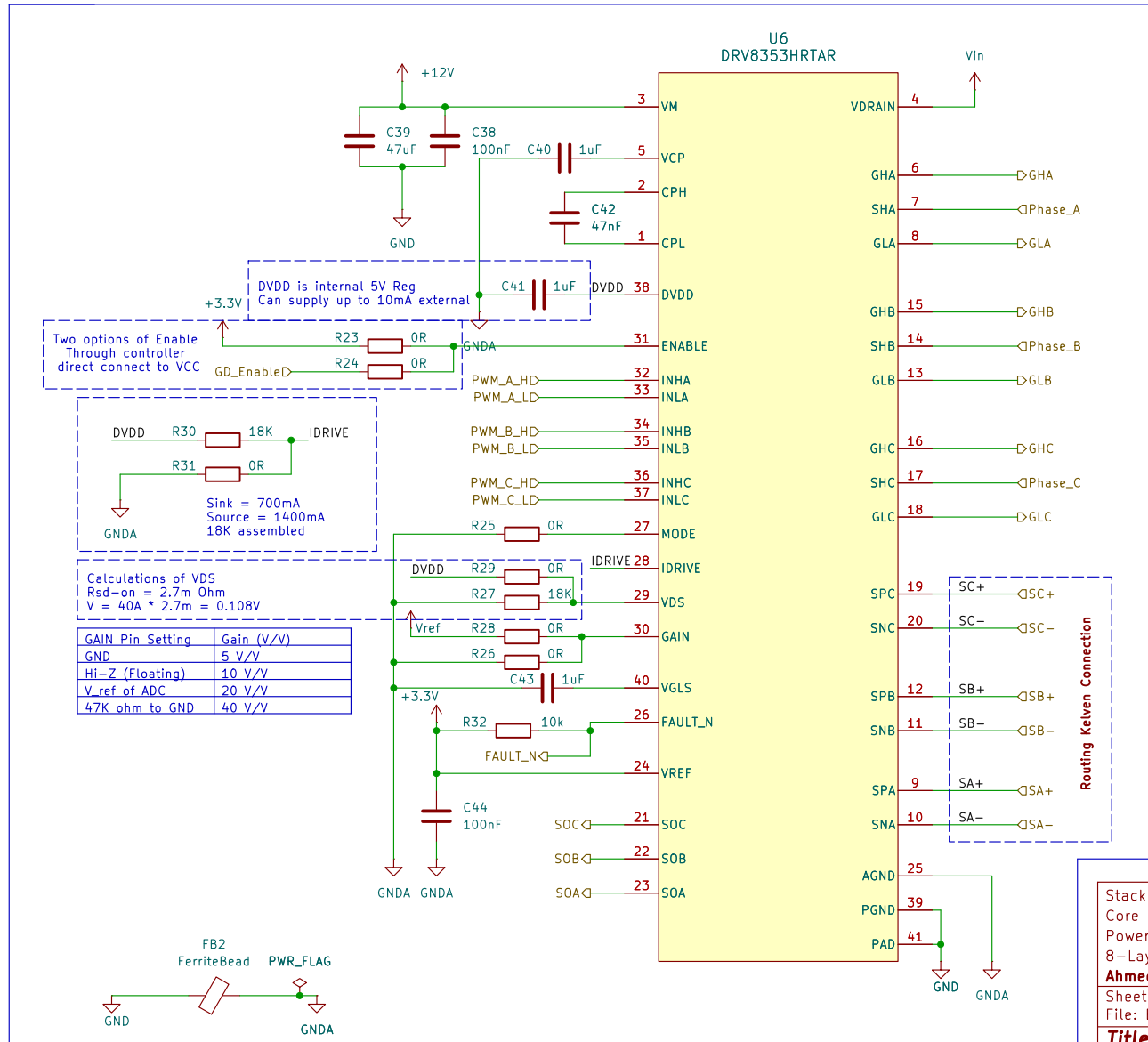
Title: BLDC_MC

Size: A4 Date: 2026-08-17
KiCad E.D.A. 10.0.5

Rev: V1.0
Id: 3/10

[4] Smart Gate Drive

for the 3 phase Inverter with current sensor



Gate Resistors vs. IDRIVE

Uses external gate resistors to limit the current charging the MOSFET gate. If you want to change the switching speed to reduce EMI or heat, you have to desolder and change the resistors.

Smart Driver: Uses an internal, programmable current source called IDRIVE. You set the source/sink current (e.g., 100mA, 1A) via a single pin or SPI. This allows you to tune the switching speed of your inverter without changing any hardware components.

Stackup & Sensing: System Ratings: 100V DC Bus | 50A Continuous | T_A = 125 degrees
Core MCU: STM32G431RBT6 (FOC & CORDIC Processing, 100 kHz PWM)
Power Stage: 3-Phase Inverter with Active Reverse Polarity & E-Fuse OCP
8-Layer High-TG180 PCB with Kelvin Current Sensing

Ahmed Aboeita

Sheet: /BLDC_MC_V1.0_Gate_Drive/
File: BLDC_MC_V1.0_Gate_Drive.kicad_sch

Title: BLDC_MC

Size: A4
KiCad E.D.A. 10.0.5

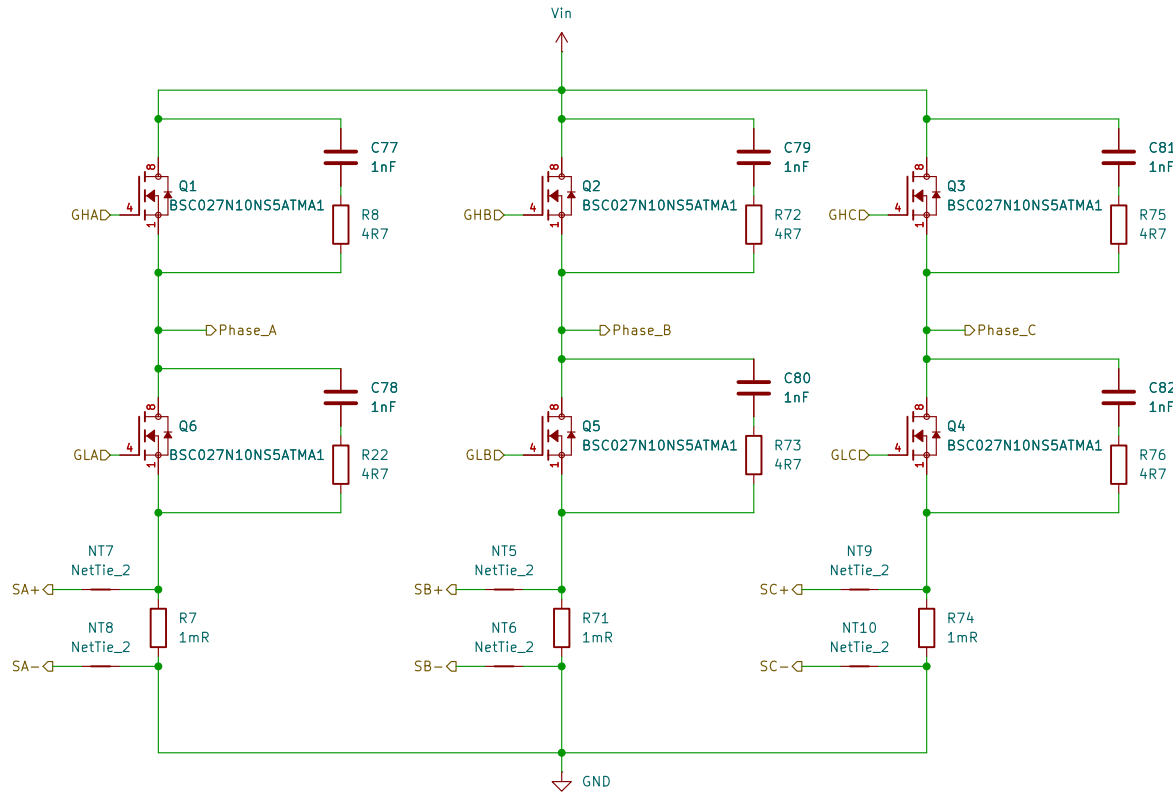
Date: 2026-08-17

Rev: V1.0

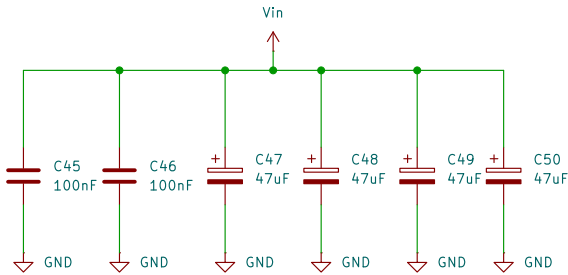
Id: 4/10

[5] 3 Phases Inverter Bridge

3 Phase Inverter Bridge



DC Link



Steps for adjusting snubber circuit

- 1- Measure the Initial Ringing
- 2- Add the "Shift" Capacitor - increase the value of C_add until the ringing frequency is exactly half of the original frequency $f_{new} = 0.5 * f_{ring}$.
- 3- Calculate the parasitic capacitance $C_p = C_{add} / 3$
- 4- Calculate Parasitic Inductance $L_p \rightarrow L_p = 1 / ((2 * \pi * f_{ring})^2 * C_p)$
- 5- Determine the Snubber Resistor $R_{snub} \rightarrow R_{snub} = \sqrt{L_p / C_p}$

Stackup & Sensing: System Ratings: 100V DC Bus | 50A Continuous | T_A = 125 degrees
Core MCU: STM32G431RBT6 (FOC & CORDIC Processing, 100 kHz PWM)
Power Stage: 3-Phase Inverter with Active Reverse Polarity & E-Fuse OCP
8-Layer High-TG180 PCB with Kelvin Current Sensing

Ahmed Aboeita

Sheet: /BLDC_MC_V1.0_3Phase_Inverter_legs/
File: BLDC_MC_V1.0_3Phase_Inverter_legs.kicad_sch

Title: BLDC_MC

Size: A4

Date: 2026-08-17

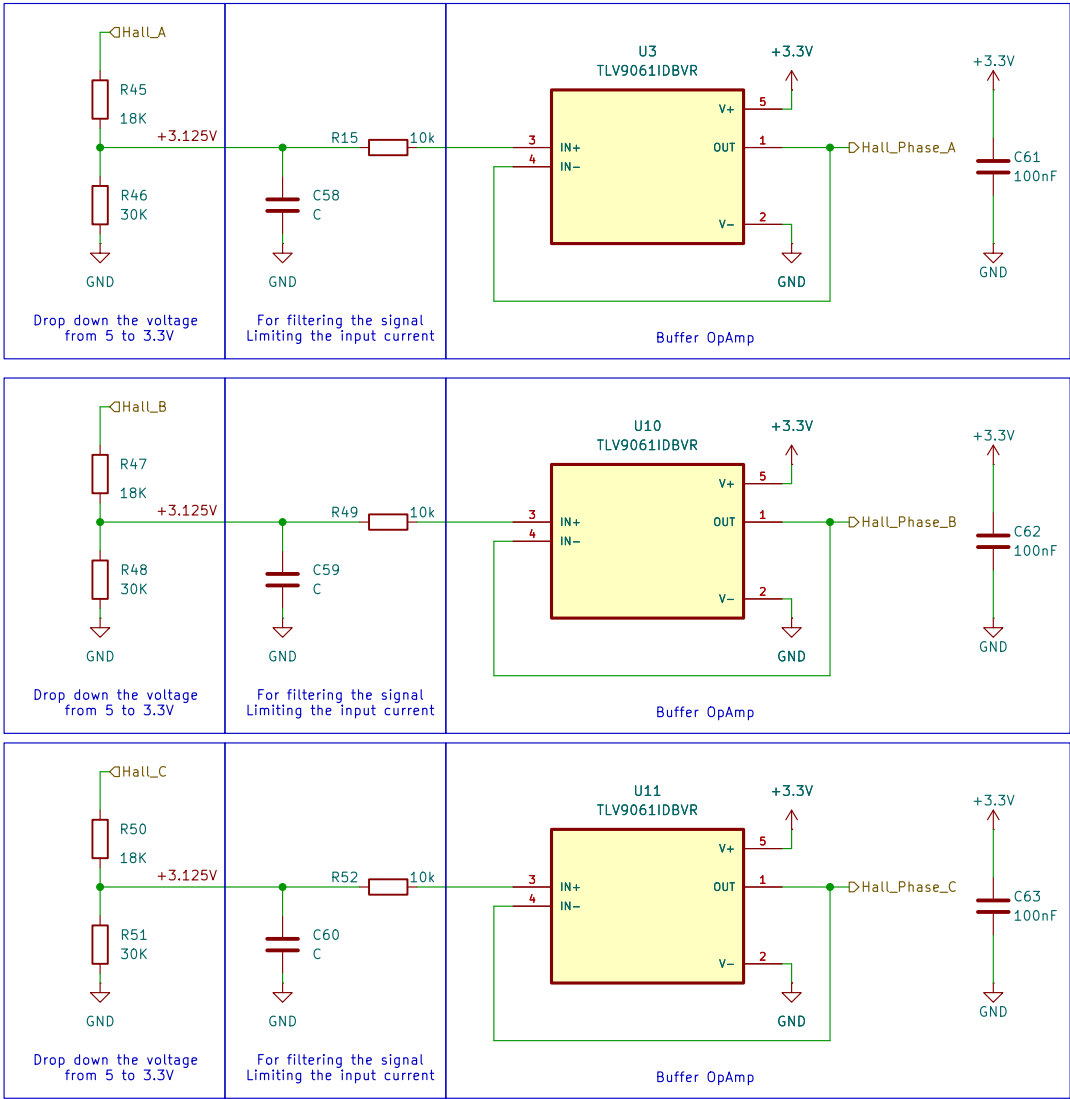
Rev: V1.0

KiCad E.D.A. 10.0.5

Id: 5/10

[6] Hall Effect Sensor Interface

For the 3 Phases of Hall Sensor



Stackup & Sensing: System Ratings: 100V DC Bus | 50A Continuous | T_A = 125 degrees
Core MCU: STM32G431RBT6 (FOC & CORDIC Processing, 100 kHz PWM)
Power Stage: 3-Phase Inverter with Active Reverse Polarity & E-Fuse OCP
8-Layer High-TG180 PCB with Kelvin Current Sensing

Ahmed Aboeita

Sheet: /BLDC_MC_V1.0_HallSensor/
File: BLDC_MC_V1.0_HallSensor.kicad_sch

Title: BLDC_MC

Size: A4

Date: 2026-08-17

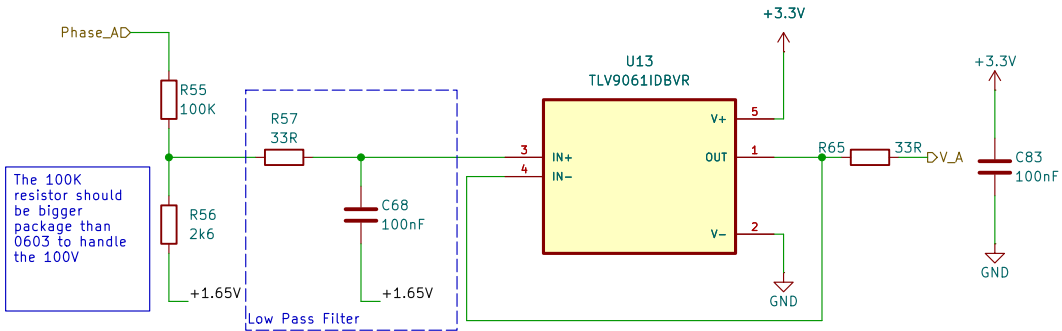
Rev: V1.0

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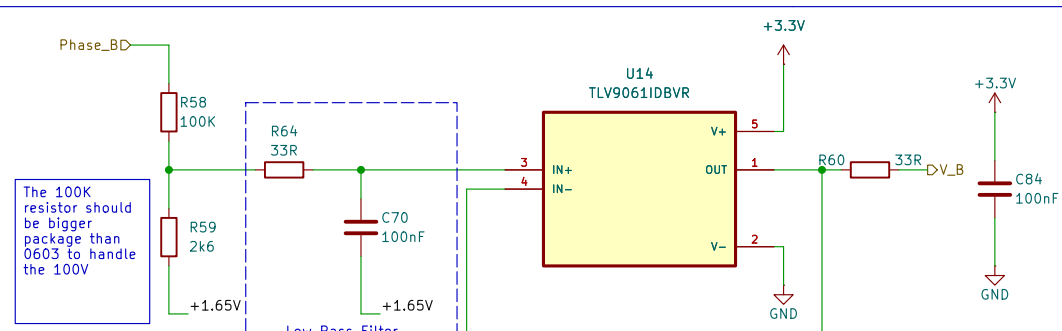
Id: 6/10

[7] 3 Phase Inverter Voltage Sensing

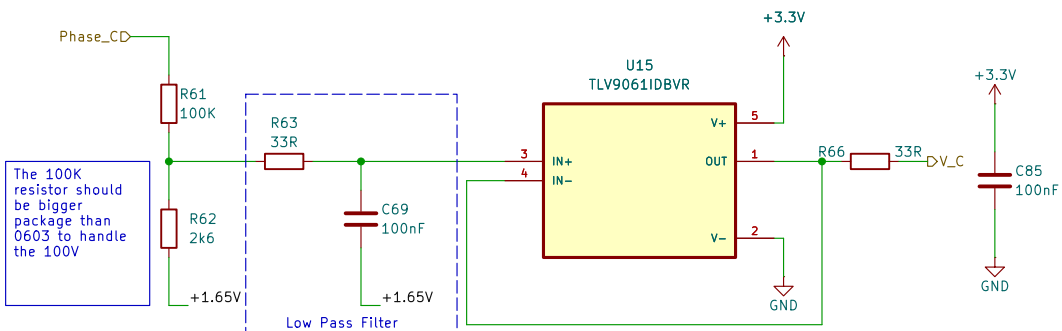
Phase A



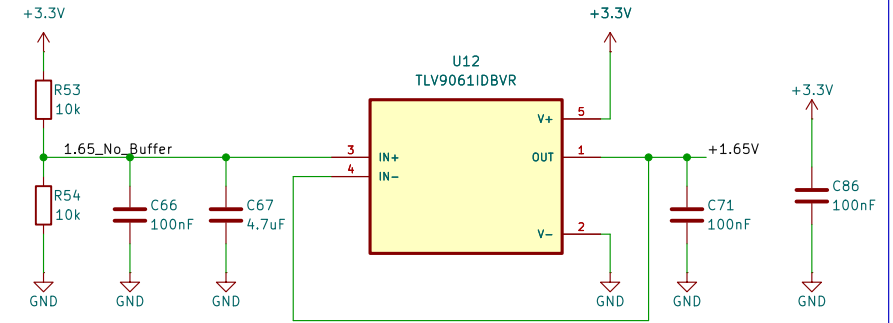
Phase B



Phase C



Virtual Ground +1.65V



make a virtual ground to read the negative voltage

Stackup & Sensing: System Ratings: 100V DC Bus | 50A Continuous | T_A = 125 degrees
Core MCU: STM32G431RBT6 (FOC & CORDIC Processing, 100 kHz PWM)
Power Stage: 3-Phase Inverter with Active Reverse Polarity & E-Fuse OCP
8-Layer High-TG180 PCB with Kelvin Current Sensing

Ahmed Aboeita

Sheet: /BLDC_MC_V1.0_Voltage_Sensor/
File: BLDC_MC_V1.0_Voltage_Sensor.kicad_sch

Title: BLDC_MC

Size: A4 Date: 2026-08-17

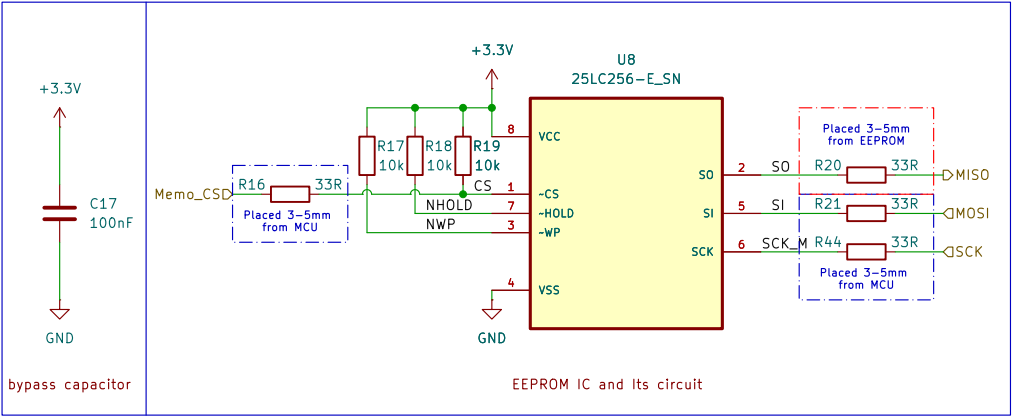
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Rev: V1.0

Id: 7/10

[8] EEPROM Memory 256Kbit

This memory used to store important data and configuration parameters



Stackup & Sensing: System Ratings: 100V DC Bus | 50A Continuous | T_A = 125 degrees
Core MCU: STM32G431RBT6 (FOC & CORDIC Processing, 100 kHz PWM)
Power Stage: 3-Phase Inverter with Active Reverse Polarity & E-Fuse OCP
8-Layer High-TG180 PCB with Kelvin Current Sensing

Ahmed Aboeita

Sheet: /BLDC_MC_V1.0_Memory/
File: BLDC_MC_V1.0_Memory.kicad_sch

Title: BLDC_MC

Size: A4

Date: 2026-08-17

Rev: V1.0

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Id: 8/10

This IC can operate as CAN-FD for faster datarate



Series Resistors Help manage signal integrity by damping potential high-frequency noise or reflections on the digital traces.

Parallel Capacitors is used to filter noise on the digital I/O lines

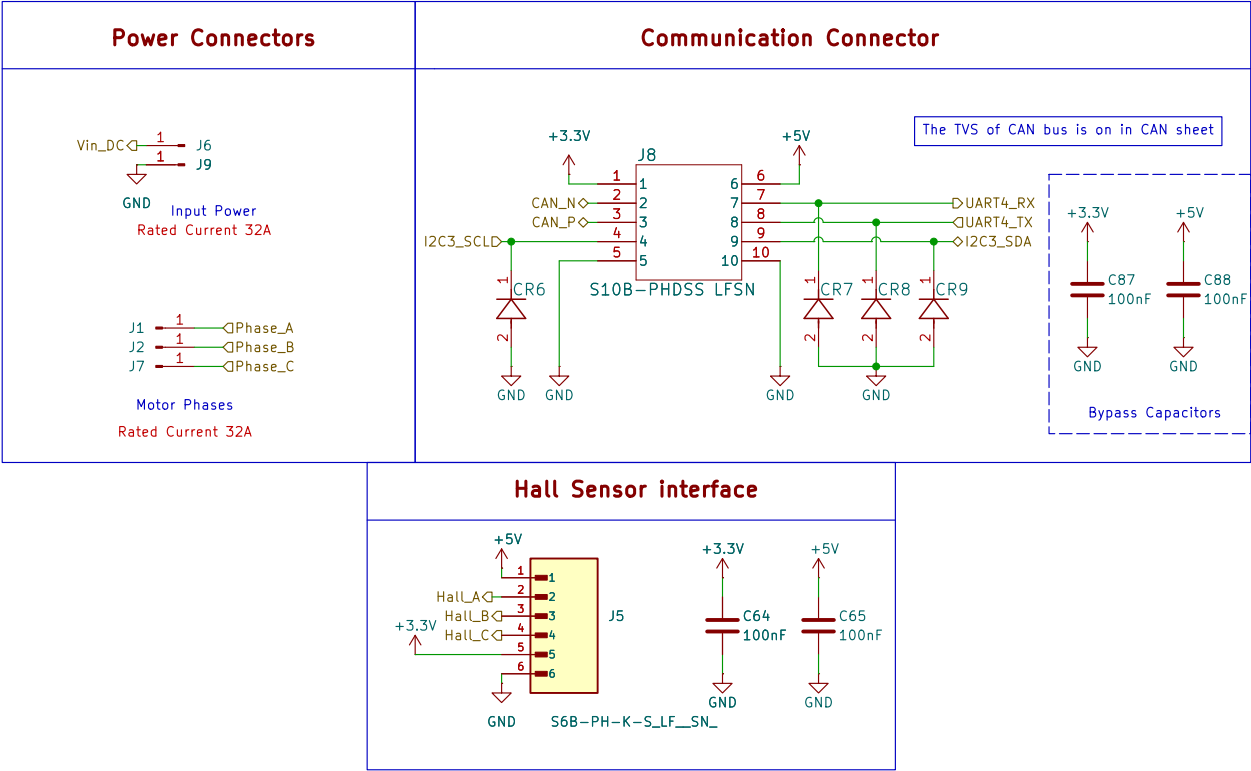
Ahmed Aboeita

Title: BLDC_MC

Rev: V1.0

Id: 9/10

[10] Board's Connectors Power & Communication



Stackup & Sensing: System Ratings: 100V DC Bus | 50A Continuous | T_A = 125 degrees
Core MCU: STM32G431RBT6 (FOC & CORDIC Processing, 100 kHz PWM)
Power Stage: 3-Phase Inverter with Active Reverse Polarity & E-Fuse OCP
8-Layer High-TG180 PCB with Kelvin Current Sensing

Ahmed Aboeita

Sheet: /BLDC_MC_V1.0_Connectors/
File: BLDC_MC_V1.0_Connectors.kicad_sch

Title: BLDC_MC

Size: A4 Date: 2026-08-17

KiCad E.D.A. 10.0.5

Rev: V1.0

Id: 10/10